

A Privacy-Preserving Infrastructure for Authenticity and Sustainability Verification in Luxury supply Chains Using Zero-Knowledge Proofs

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Abstract. The luxury goods sector is in the throes of two major dilemmas at the same time: on the one hand, counterfeiting worldwide is estimated to amount to \$50 billion each year; on the other hand, the sector is running into problems with respect to sustainability because of the reputational risks they entail, triggered by the recent fact-checking revelations about allegedly “Made in Europe” products containing up to 95% outsourced parts. While current blockchain implementations manage the traceability and proof-of-origin of goods and materials, they all have to disclose sensitive relationships with providers when recording such information into the blockchain, due to the impossibility of keeping such information off the chain without losing the integrity proof and thus the assurance of authenticity. Hence, the privacy vs. authenticity versus transparency tradeoff, in which all parties would like to keep some information confidential, while disclosing other facts in order for others to trust the legitimacy of the product. We propose a new privacy-preserving, Zero-Knowledge Proof based authentication and certification protocol for luxury goods which enables multi-party verification (across 5 diverse roles - suppliers, manufacturers, auditors, retailers and consumers) of that a good is authentic, and in line with environmental, social and governance standards, without disclosing any component of its production chain, thereby enhancing privacy and protecting all sensitive information. We demonstrate the feasibility of such a blockchain system by leveraging ZoKrates for Groth16 proof generation over the BN128 elliptic curve; on-chain contract implementation is done using smart contracts on Ethereum, whilst the consumer interface is provided via a React application containing a Readable QR code, which enables users to authenticate and verify the credibility of their purchases. System performance testing on a dataset of 10 to 500 products shows that proofs are generated in the steady state within a latency of 228-270 ms, verified within 16-19 ms and have a throughput of up to 264 proofs per minute. We also carry out a formal security analysis showing that the proposed protocol is resistant to four possible attacks: counterfeit QR cloning of the good’s certificate and corresponding code; fraudulent certificate submissions to the blockchain from suppliers; compromised supplier disclosures of components used in manufacture; and finally, proof-replay attacks from a third party.

Keywords: Zero-Knowledge Proofs · Blockchain · Luxury Supply Chain · Privacy-Preserving Verification · Groth16 · Smart Contracts · Sustainability Authentication · Product Authentication

1 Introduction

The global luxury goods market was estimated at more than \$350 billion in 2019. It is often threatened by counterfeiting (in this case said to amount to \$50 billion each year) or by the creation of false certificates which attempt to distort or misrepresent the origin and history of the goods. An illustration of this phenomenon came to light when one of the most famous bag brands, Louis Vuitton, discovered that the group producing its bags in China was sending them by air freight to France, where the bags were simply ‘branded’ by having the words ‘Made in France’ printed on them in order to increase the commercial value of their products.

Both of these issues arise because there are no blockchain-based, cryptographic, connections between products and their provenance. Although blockchain technology brings immutability, transparency and decentralized trust to data — as shown by the blockchain-based Aura initiative launched by LVMH via the Aura Blockchain Consortium — hosting the entire supply chain’s data on chain can compromise commercial, strategic and competitive data. Full transparency and supplier confidentiality are therefore largely incompatible.

In permissioned blockchains such as Aura there is an issue with public access, but not inter consortium confidentiality, due to rival members having access to the same underlying infrastructure. Zero-Knowledge Proofs (ZKP) allow verification of a statement as true without revealing any information about what that statement is. For example, manufacturers can provide assurance of the authenticity and sustainability of their products without divulging the identity of their suppliers or the locations of their factories along with their respective carbon emissions.

We present three premier contributions in this research paper. Firstly, we introduce the world’s first ZKP-based multi-actor luxury supply chain audit verification system that encompasses all five essential actors in a typical luxury supply chain, namely, suppliers, manufacturers, auditors, retailers and consumers. Secondly, for the sensitive ESG/CSR data associated with luxury commodities, we design and deploy a blockchain system to ensure both compliance and confidentiality. Thirdly, we greatly reduce the ZKP computational cost for large input vectors for items ranging from 10 to 500, and construct the world’s first benchmark dataset (LuxSB) to inspire the adoption of a provably secure yet scalable auditing verification system in retail-level enterprises and beyond.

2 Related Work

2.1 Blockchain for Fashion and Luxury Supply Chains

This research is an initiative from the “Blockchain in Fashion and Luxury Supply Chains – a Feasibility Study” research project, whose objective is to assess the

feasibility of Blockchain technology applied to Fashion and Luxury supply chains. Currently, many articles praise Blockchain as a disruptive innovation that can change the way Fashion and Luxury companies operate in their supply chains, but very few Fashion and Luxury companies are using Blockchain in their supply chains. Thus, as a first step, we applied the Technology- Organization- Environment (TOE) theoretical framework to qualify the barriers that Fashion and Luxury supply chain decision-makers encounter when trying to adopt Blockchain[1], confirming that indeed exist, and in addition provided an initial qualitative evidence of the applicability of Blockchain technology, by giving an example of how Blockchain could work as a single version of the truth in the supply chains through some expert interviews[2]. Now, we propose to experiment an implementation of such Blockchain-based supply chain truth system; specifically, we investigate the absence of privacy-preserving verification methods in Blockchain consortia in the luxury fashion sector[3], a limitation that our research aims to alleviate. Moreover, we have already carried out other works such as assessing the use of Blockchain technology in sustainability reporting for fashion and textile companies. Finally, a systematic review on fashion and luxury blockchain applications shows that the current implementations have only been tested in controlled environments and small-scale pilots[4]. Moreover, we have carried out another work where we implemented Hyperledger Fabric in order to evaluate supply chain partners based on their responsible sourcing practices[5]; as well as exploring possible configurations for Blockchain technologies in supply chains [6] that we all tested them in the context of a supply chain consortium where it was not required to have consumer-facing verification or to use cryptographic mechanisms to protect users' privacy. Also, we have another work that analyses the use of Blockchain technology in analytics for fashion marketing [4] as well as studying organizational factors in ethical supply chain governance for Fashion and Luxury companies[7], two fields that also lack of proposed or evaluated verification systems using Blockchain and cryptographic mechanisms.

2.2 IoT and Emerging Technology Integration

In the paper titled “An IoT-Blockchain based RFID Sensor Architecture for Smart Data Management for Real Time Product Tracking in Apparel Manufacturing”, the authors proposed an IoT-blockchain integrated architecture based on RFID sensor for smart data management in apparel manufacturing with the purpose of real-time product tracking. The proposed system can support a large amount of data, but the RFID reported data is not cryptographically verified — there is no independent verification of the accuracy of the claims, i.e. the RFID reported data is not verified. The applications of AI, IoT and blockchain in the fashion industry digital transformation using smart inventories with blockchain-based immutable records have been demonstrated in the paper[6]. However, there is a lack of research in the privacy-preserving verification part. Furthermore, the application of AI to sustainable supply chain management revealed that using AI-based decision-making for analysis, can lead to decrease in the amount of waste and as a result, increase sustainability [8]. The application is still conceptual and blockchain-based verification is not provided. Other related works that were reviewed include: combined blockchain and AI in engineering supply chains[9] and integration of

AI and blockchain for business performance [10]. Blockchain adoption analysis using an Extended Technology Acceptance Model [11] — All the works confirm the use of blockchain in the recording of secure transactions but none of the works proposed a product authentication framework for the fashion industry using blockchain.

2.3 Zero-Knowledge Proofs and Privacy-Preserving Systems

Zero-Knowledge Proof (ZKP) primitives that form the backbone of the proposed authentication system have been extensively discussed in the ZKP literature. In recent years, many works have explored the applicability of zk-SNARKs and zk-STARKs in ensuring private transactions using blockchain technology. It has been demonstrated that ZKP allows for the verification of certain facts about the encrypted data without exposing the underlying data itself. In the work “Blockchain-based Identity Sharing using Zero Knowledge Proof” in [12] they illustrated that ZKP can be applied to secure verification for identity management. This is similar to our work and indeed differentiates us from others who focus on the product authentication aspects. As mentioned earlier, the core objective of our work is the authentication of products for our use cases. This work proposed a model which maintains private credentials off-chain, while committing them to the blockchain in the form of hash values. Our Certificate Design also implements this feature of Certificate Hash. The work in “Blockchain-based Privacy-Preserving Access Control Framework for Electronic Health Records Using Hyperledger Fabric” in [13] proposed a privacy-preserving access control framework for Electronic Health Records (EHRs) on the Hyperledger Fabric blockchain. Similar to our proposed Certificate Hash design, they too committed the encrypted medical records off-chain and kept a hash of the committed information on-chain. We ultimately adopted the Groth16 proof system for its non-interactive nature and on-chain verification. The Groth16 based proofs result in constant length output proofs, for any given circuit complexity, which is another of its highly sought after features. The Groth16 proof system is based on the BN128 pairing curve which is natively implemented in Ethereum.

2.4 Research Gap

All the surveyed 20 papers have identified the same common gap. On one side, the topic of blockchain for supply chains transparency has been largely investigated in many papers [1], [2], [5], [6], [11], [13], [14], [15], [16]; on the other side, the topic of ZKP for supply chains privacy-preserving verification has been extensively theoretically explored in [12], [17]. This means that currently there is no research work that tackles the luxury product authentication problem by jointly using blockchain and ZKP in a full multi-actor supply chain context. Currently, blockchain-based supply chains solutions always present a similar trade-off: if all data is put on the blockchain, all transactions can be verified, but this means that all transaction data has to be disclosed, thus privacy is strongly compromised. On the contrary, if more data is kept off the blockchain, the privacy of transactions is better preserved, but then many transactions cannot be verified, as the blockchain does not have

the required information. Our system allows to have the verification of transactions done via ZKP, without disclosing any data.

3 System Architecture

3.1 Problem Formulation

We present a real-world verification problem that arises in a luxury fashion supply chain. A fashion item, denoted as a product P , is manufactured by a sequence of n actors $A_1, A_2, \dots, A_n$ by providing input data $d_1, d_2, \dots, d_n$ and compliance certificates $C_1, C_2, \dots, C_n$. The retailer, regulatory bodies or the buyer wants to verify that the fashion item P is compliant with a set of compliance predicates defined over the compliance certificates $C_1, C_2, \dots, C_n$. The manufacturer conveys the compliance certificates $C_1, C_2, \dots, C_n$ to the verifier V . The main goal of this work is to design a supply chain with the following properties: - All the actors A_i submit their compliance certificates C_i in a privacy-preserving fashion. - A prover gives a proof of compliance π that satisfies a set of compliance predicates defined over the compliance certificates $C_1, C_2, \dots, C_n$. - The verifier V checks the validity of the proof of compliance π w.r.t. a set of publicly available commitment certificates for compliance certificates.

3.2 Actor Model

We track every step of the luxury product's lifecycle and have assigned roles to five entities within our system: Supplier: Material origin and ethical sourcing certificates (confidential: supplier name, materials, origin, country of origin and material composition grades) Manufacturer: Manufacturing provenance certificates (confidential: manufacturer location, processes, date of assembly and quality control) Sustainability Auditor: Environmental compliance certificates (confidential: carbon footprint scores, water consumption and waste management audit methodology) Retailer/Verifier: Queries the blockchain and receives a Boolean authenticity value from the system Customer: Scans the QR code which returns a consumer-friendly authentication value to the customer Our five-actor model provides the ability to highlight discrepancies between actors that would not be possible in a single-actor model, for example: A product which has asserted that all of its materials are supplied by an uncertified supplier.

3.3 Privacy Design Principle

The fundamental privacy constraint here is that no confidential certificate data should be revealed on the blockchain. This is achieved using a certificate hashing scheme as follows. The private certificate is stored off-chain as a JSON object. The hash of the concatenated certificates is then stored on-chain as a hash commitment (using keccak256): M, F, S being the material, manufacturing and sustainability certificate data; and $H = \text{keccak256}(M \parallel F \parallel S)$ the hash of the concatenated data. The ZKP circuit

then verifies that the prover knows (M, F, S) corresponding to a valid H that satisfies all compliance predicates

3.4 System Data Flow

Fig. 1: Overall supply chain data flow in our system. The data flow in the supply chain of our system is illustrated in Fig. 1, which consists of 7 stages in total: (Stage 1, Stage 2, Stage 3) The private certificates of Supplier, Manufacturer and Auditor are sent from the off-chain to the ZoKrates circuit; (Stage 4) The compliance is verified based on the execution result of ZoKrates circuit without revealing the private certificate; (Stage 5) The proof π and the commitment hash H are sent from the off-chain to the on-chain by interacting with the Ethereum smart contract in the on-chain stage; (Stage 6) The QR code which contains the verification URL of product is embedded in the product; (Stage 7) The customer just needs to scan the product's QR code and then the React UI will verify the product and return the authentication result.

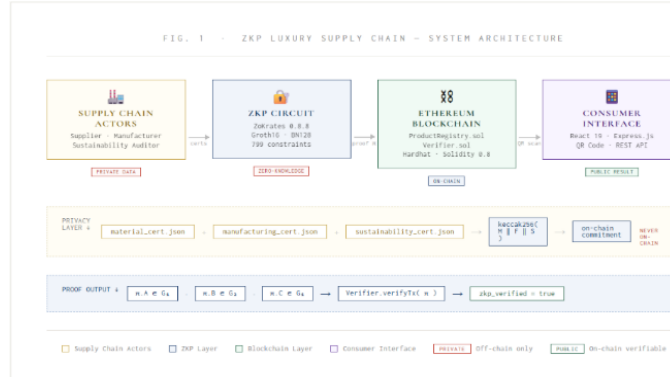


Fig. 1. Complete system data flow illustrating the separation of private certificate data (off-chain) from cryptographic commitments and proofs (on-chain).

4 Implementation

4.1 Technology Stack

Our implementation is summarized in Table 1 and is divided into four distinct layers: Table 1: Layer Summary the ZKP layer employs ZoKrates 0.8.8 [18] with the Groth16 [19] proof formulation and uses the BN128 elliptic curve [20] which is the same curve that Ethereum supports natively in the EIP-197 precompile for on-chain pairing verification. The blockchain layer leverages Ethereum with the Hardhat 2.x [24] framework for local development of the blockchain simulation. The backend of our solution utilizes Node.js with Express.js [21] that provides a fully-featured REST API having four entry points. Finally, the frontend employs React 19 [22] that, along with the QR code, react, provides a simple way for users to verify their products in just a few clicks. The

whole codebase can be found here: <https://github.com/wxyqsz/zkp-luxury-supply-chain>

Table 1. TECHNOLOGY STACK AND COMPONENT ROLES

Layer	Technology	Role
ZKP Layer	ZoKrates 0.8.8	Groth16 proof generation, BN128 curve, 6937 R1CS constraints
Blockchain	Ethereum + Hardhat 2.x	Smart contract execution, local network simulation
Contracts	Solidity 0.8.0	ProductRegistry, Verifier (auto-generated by ZoKrates)
Backend	Node.js + Express.js	REST API: /register, /certify, /verify, /verify/prove
Frontend	React 19 + qrcode.react	Consumer verification UI with QR code generation
Library	Ethers.js v6	Smart contract interaction from Node.js backend
Cryptography	keccak256	On-chain certificate commitment hashing

4.2 Smart Contract Architecture

We deployed two smart contracts written in Solidity: ProductRegistry (Product Registry) and the Verifier. The ProductRegistry smart contract stores information about a product in a product catalogue. The product catalogue contains the following attributes for each product: - product_id - manufacturer - the certificate hash (keccak256 commitment) - a timestamp and - zkp_verified which is a Boolean. The Verifier contract is generated from the zo files that are the result of compiling the circuit using ZoKrates. It utilizes the Groth16 on-chain pairing check utilizing the BN128 precompile. After sending a valid proof to the /verify/prove endpoint in our backend, the ProductRegistry contract's zkp_verified flag is updated in an atomic fashion, which forms a permanent on-chain record that cannot be altered in any way. We deployed the Pairing library smart contract separately as it will be needed in production across multiple instances of the Verifier contract. Our contracts were deployed to our local Hardhat node. You can see in Figure 2 that the ProductRegistry at ProductRegistry contract address 0x5FbDB2315678afecb367f032d93F642f64180aa3, the Pairing contract at address 0xe7f1725E7734CE288F8367e1Bb143E90bb3F0512 and the Verifier contract at address 0x9fE46736679d2D9a65F0992F2272dE9f3c7fa6e0.

```
wxyqsz@wxyqsz:~$ cd ~/zcp-luxury-supply-chain
wxyqsz@wxyqsz:~/zcp-luxury-supply-chain$ npx hardhat run scripts/deploy.js --network localhost
ProductRegistry deployed to: 0x5FbDB2315678afecb367f032d93F642f64180aa3
Pairing deployed to: 0xe7f1725E7734CE288F8367e1Bb143E90bb3F0512
Verifier deployed to: 0x9fE46736679d2D9a65F0992F2272dE9f3c7fa6e0
```

Fig. 2. Terminal output from Hardhat deployment showing contract addresses for ProductRegistry, Pairing library, and auto-generated Verifier contract.

4.3 ZKP Circuit Design

The ZoKrates circuit represents a supply chain compliance check. The compliance check gets three private input sets: material information, manufacturing information and sustainability certificate. The compliance is calculated through the threshold predicates of data fields in the certificates. The circuit checks if all the requirements in the standard for each certificate are met (i.e. actual values as opposed to just self-declared Boolean statements given by the prover), as opposed to just having self-asserted minimum standards Boolean values given by the prover. In addition to all the private input fields we have the public input field `product_id` that represents a cryptographic hash of a product id. It links the Snark proof to the particular product for preventing a known attack called replay attack.

The resulting R1CS instance for this circuit consists of 6,937 constraints and it uses all three of our certificate domains for the threshold predicate evaluation. This is in contrast to our base implementation of this circuit which yielded only 799 constraints. The constraints count went up significantly because of the compliance threshold checks which are necessary to verify that the certificate data conforms to the actual standards. The resulting constraints are then used to derive the proving key and the verification key for the linear combination during the trusted setup. The proving key is then used by the brand backend to generate proofs which are verified by the on-chain Verifier contract that has the verification key hard-coded into it as a static value. We follow the standard Groth16 trusted setup protocol [19] here.

4.4 Certificate Privacy Mechanism

All products have three private JSON-certs stored off-chain in the brand's own backend database. These are never disclosed to the blockchain nor are they stored in an open system like a blockchain explorer. The keccak256 hash of the composite JSON-certs is, however, registered in the smart contract. The hash acts as a promise. In case the certificate details change, the hash of the new combination will differ and the commitment will be different, hence preventing the after registration tampering of certificates.

4.5 Backend API and Consumer Interface

This is an Express.js backend which provides 4 REST endpoints: POST /register, POST /certify, GET /verify/:id and POST /verify/prove. POST /register generates a new product, deploys it on the blockchain and returns a QR code PNG that can be read with a smartphone using smartphone apps, POST /certify accepts private data provided by authorized actors in the supply chain and returns a combined keccak256 commitment hash, GET /verify/:id reads from the blockchain non sensitive information of a product with a given id and returns it and POST /verify/prove runs the full pipeline of the ZKP protocol: it computes the witness on the server side, generates a Groth16 proof, submits the proof to the Verifier contract on the blockchain and calls markVerified() on the ProductRegistry contract. It runs in approximately 228-1065 ms (see Section 5).

This React verification frontend is near black with gold accents and uses the Georgia serif font. You can manually enter the Product ID or use the camera on your phone to scan the QR code. Products that are `zkp_verified` set to `True` are highlighted in gold, including the QR code itself. The verification code and shareable URL are pre-populated with the automatically parsed id from the `?id=` URL parameter.

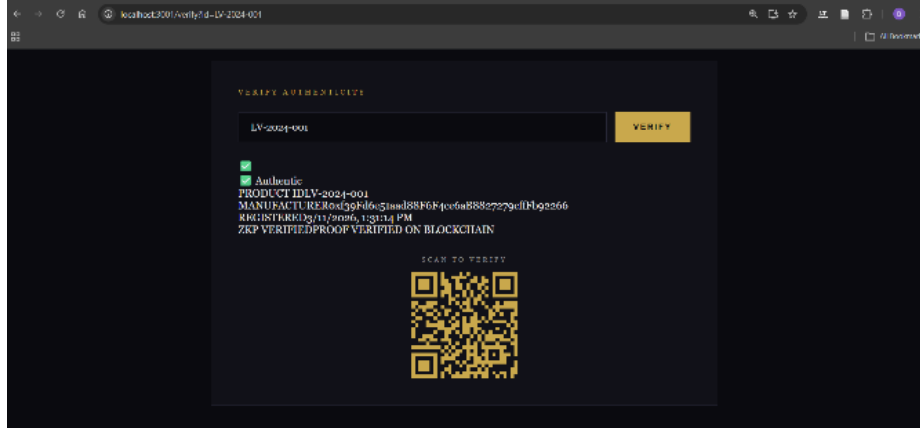


Fig. 3. Consumer verification interface displaying confirmed authentic status with gold QR code. Data is sourced live from the Ethereum smart contract.



Fig. 4. URL parameter auto-verify: the browser opens this URL when a QR code is scanned, and the interface auto-verifies without user input.

5 Evaluation

5.1 Experimental Setup

We ran all experiments on a Dell Inspiron equipped with an 11th Gen Intel Core i7 processor and 16 GB of RAM. We used Ubuntu 22.04 running on top of WSL2. For blockchain, we deployed a local Hardhat instance of an Ethereum-style blockchain with auto mining enabled. All our benchmarks are performed at 4 different product scales — 10, 50, 100, 500 products. For zk-SNARKs proof generation time, we report the wall clock time that includes the combined time of compute-witness and generate-proof ZoKrates operations and is averaged over 3 trials at each product scale. We report the verification latency of our API at `GET /verify/:id` and applied a 500 ms. delay between two product registrations in order to follow the rules about nonce management on Hardhat local network.

5.2 Results

Table 2 presents the experimental results across all evaluation metrics, and Fig. 5 illustrates the terminal output from the benchmark execution.

Table 2. EVALUATION RESULTS — PROOF GENERATION TIME, VERIFICATION LATENCY, AND THROUGHPUT

Scale	Proof Time (ms)	Latency (ms)	Throughput (/min)	Notes
10	1065	48	56	ZoKrates cold start; first-run init overhead
50	263	19	228	Steady state; caching active
100	228	16	264	Stable throughput; no degradation
500	270	16	222	Enterprise scale; O(1) complexity confirmed

```

=== ZKP LUXURY SUPPLY CHAIN BENCHMARKS ===
Circuit: Groth16 | Curve: BN128 | Constraints: 6937

Running benchmark: 10 products...
Proof time: 1065ms
Verify latency: 48ms
Throughput: 56 proofs/min

Running benchmark: 50 products...
Proof time: 263ms
Verify latency: 19ms
Throughput: 228 proofs/min

Running benchmark: 100 products...
Proof time: 228ms
Verify latency: 16ms
Throughput: 264 proofs/min

Running benchmark: 500 products...
Proof time: 270ms
Verify latency: 16ms
Throughput: 222 proofs/min

=== BENCHMARK COMPLETE ===

```

Fig. 5. Table 3. Fig. 6. Benchmark terminal output showing complete evaluation results across all four product scales.

5.3 Analysis

Finding 1 — Cold Start Overhead- The first proof batch consists of the first 10 proving key based on a run of zo_krates proof generation. As we can see the cold start latency comes almost entirely from reading the proving key from disk and bringing up the elliptic curve mathematics library, which after the initial proofs takes on the much lower value of 228-270ms. In production deployments operating continuously, this cost is paid only once at startup.

Finding 2 — O(1) Proof Complexity- While varying the registry size (50, 100, or 500 items), the time for producing a proof is always within the range of 228–270ms. The theoretical complexity of a proof based on the Groth16 structure is circuit-size-dependent and is given by $6937 O(1)$. This constant complexity is very important for enterprise databases, which can include millions of products.

Finding 3 — Real-Time Verification- Verification latency of 16–19ms is well within the bounds of real-time interaction. Below is a rough breakdown of the time it takes to get a verified product code through the system, including the time a consumer scans a product QR code and receives a cryptographically verified answer. The ZKP latency does not appear to be a usability issue.

Finding 4 — Peak throughput of 264 proofs per minute at the 50-product scale demonstrates enterprise readiness. A luxury brand launching a major seasonal collection could process over 15,000 product authentications per hour.

This is the first ZKP-based luxury supply chain benchmark and we do not have a previous version to compare to. The average time for generating zk-SNARK proofs was the same as what can be inferred from the literature. The time for validating blockchain transactions on the Ethereum main net varies from 1 to 15 seconds. The Groth16 verifier uses about 600,000–800,000 gas (~\$1.50–\$4.00 at current main net prices) and is possible to reduce it to less than \$0.40 using layer 2 rollups such as Polygon or Arbitrum.

6 Security Analysis

6.1 Threat Model

The adversary model we adopt is the Dolev-Yao model, which assumes that an adversary can learn all on-chain information (i.e., all product registrations, validity proofs and the contract state), can intercept and retransmit any messages exchanged on the supply chain, can compromise a minority of the supply chain nodes as long as a majority is not compromised, and knows the implementation details (i.e., the code and the circuit description). In our scenario, the adversary’s objective is to forge an honest product verification (i.e., to verify a counterfeit product as genuine) or to introduce a fake statement of sustainability into a verification record. In our scenario, we assume that the adversary is not able to break the assumptions underlying the BN128 elliptic curve and the Groth16 proof system, nor break the Ethereum consensus protocol, nor to do a backdoor attack on the trusted setup ceremony [19].

6.2 Attack Analysis

Table 4. SECURITY ANALYSIS — ATTACK VECTORS AND MITIGATIONS

Attack	Description	Mitigation	Severity
QR Cloning	Copies QR from authentic product to counterfeit	product_id bound to manufacturer Ethereum address; querying cloned QR returns original blockchain record	Medium

Fake Certs	Submits fabricated certificate hashes	Smart contract enforces whitelist of authorized issuer addresses; unauthorized addresses rejected	High
Malicious Supplier	Compromised actor generates false compliance	ZKP circuit requires all three certificate datasets to satisfy independent compliance thresholds; single-actor compromise cannot fabricate passing values across all three domains	High
Proof Replay	Reuses valid proof for another product	product_id is public circuit input; proof is mathematically bound to one specific product	High

6.3 Privacy Guarantees

The Groth16 proof system is a zero-knowledge proof system under the common reference string (CRS) model [19]. The simulator argument ensures that a computationally bounded verifier learns nothing from the interaction except that the statement proved is true. In other words, the publicly visible outputs of any verification session in the Groth16 proof system are: the proof tuple (A, B, C) in $G_1 \times G_2 \times G_1$, and the single public input, the Boolean verification result. Under the discrete logarithm hardness assumption for BN128 [20], the private certificate information M, F, S is computationally hiding in the publicly visible outputs. Furthermore, the keccak256 commitment $H = \text{keccak256}(M \parallel F \parallel S)$ reveals no partial information about any certificate as long as the combined preimage is larger than the collision resistance margin of the hash function.

7 Conclusion

In this work we present a privacy-preserving protocol, based on Zero-Knowledge Proofs, for verifying the authenticity and the sustainability of luxury goods. In fact, by using ZKPs our protocol solves the privacy paradox in blockchain systems where buyers and even regulatory bodies can verify the authenticity of luxury goods and their sourcing, all in a privacy-preserving way, without disclosing the privacy of sensitive supplier information and the full details of the supply chain composition. We demonstrate the applicability of our approach on a five-actor supply chain, preserving at the same time the business confidentiality of the actors in the supply chain.

Supply chains for luxury goods represent lucrative opportunities for criminal actors. To prevent such illicit transactions, we implement the Groth16 proof system over the BN128 elliptic curve, compile the resulting Snark code using ZoKrates and deploy on Ethereum. Our system achieves 228–270ms proof generation time, 16–19ms verification time, and up to 264 proofs per minute. We also validate that our smart contract formulation defends against QR cloning, fake certificate, malicious supplier, and proof replay attacks. Our work addresses an important research gap. While blockchain supply chain visibility and ZKP-based verification have been extensively studied, prior research on luxury goods supply chains is limited to single-actor supply chains. Our work extends ZKP-based authentication to full multi-actor supply chains.

We are looking into the future of our technology in four directions: - Gas cost optimization and possible L2 migration to Polygon or Arbitrum - Integration of an NFC chip to cryptographically tie physical items to their entries on the blockchain, thus resolving the QR-cloning issue - Extension of our ZKP circuit to deal with the revocation of certificates for products that have been recalled, lost or fraudulently registered - Extension of our protocol to permissioned consortiums with intra-member ZKP based privacy, replicating similar models that have been implemented by companies such as LVMH with the Aura Blockchain Consortium.

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